

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5) **Assessment Tool Weightage: 40%**

QUESTIONS:5

Total Marks 50

Duration :60 Minutes Annexure A

Industry Problem Solving Task

Code of Conduct:

I K.Mounith(192425090)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.mounith

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	

Professionalism	5%	Highly professional, welldocumented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	
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Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Traffic Surveillance Model

Selection A traffic monitoring system must:

- Accuracy $\geq 90\%$
- Latency ≤ 80 ms
- Works under varying lighting Options:

Method A: 88%, 60 ms

Method B: 91%, 85 ms

Method C: 92%, 75 ms

Compare and evaluate which method satisfies all constraints. Justify logically.

A. Traffic Surveillance Model Selection

Problem:

A traffic surveillance system requires an accuracy of at least 90%, latency of not more than 80 ms, and reliable performance under varying lighting conditions. Three methods are available: Method A with 88% accuracy and 60 ms latency, Method B with 91% accuracy and 85 ms latency, and Method C with 92% accuracy and 75 ms latency.

Comparison and Analysis:

Method A has a latency of 60 ms, which satisfies the latency requirement. However, its accuracy is only 88%, which is below the required 90%. Therefore, Method A is not suitable.

Method B provides 91% accuracy, satisfying the accuracy requirement. However, its latency is 85 ms, which exceeds the maximum allowed latency of 80 ms. Therefore, Method B is also not suitable.

Method C provides 92% accuracy and 75 ms latency. Its accuracy is above the required 90%, and its latency is below the maximum limit of 80 ms. Therefore, Method C satisfies both major performance requirements.

Engineering Justification:

For a real-time traffic surveillance system, both accuracy and low latency are important. Method C provides a good balance between these two factors. It can provide accurate traffic detection while maintaining an acceptable response time. Its performance should also

be tested under different lighting conditions such as daylight, nighttime, shadows, and glare.

Final Solution:

Method C should be selected for the traffic surveillance system because it is the only method that satisfies both the accuracy and latency requirements.

Accuracy = 92% $\geq$ 90%

Latency = 75 ms $\leq$ 80 ms

Conclusion:

Method C is the most suitable and practical solution because it provides high accuracy, low latency, and better suitability for real-time traffic surveillance.

2. Drone Vision System Constraint A drone system requires:

- Lightweight computation
- Real-time detection (<60 ms)
- Accuracy $\geq$ 85% Options:

Model A: 83%, 40 ms

Model B: 87%, 65 ms

Model C: 86%, 55 ms

Compare and evaluate the suitable model. Justify using constraints.

A. Drone Vision System Constraint

Problem:

A drone vision system requires lightweight computation, real-time detection with latency less than 60 ms, and accuracy of at least 85%.

Three models are available: Model A with 83% accuracy and 40 ms latency, Model B with 87% accuracy and 65 ms latency, and Model C with 86% accuracy and 55 ms latency.

Comparison and Analysis:

Model A has a latency of 40 ms, which satisfies the real-time requirement. However, its accuracy is only 83%, which is below the required 85%. Therefore, Model A does not satisfy all the system requirements.

Model B provides an accuracy of 87%, which satisfies the accuracy requirement. However, its latency is 65 ms, which is greater than the required limit of 60 ms. Therefore, Model B is not suitable for real-time drone detection.

Model C provides 86% accuracy and 55 ms latency. Its accuracy is above the required 85%, and its latency is below 60 ms. Therefore, Model C satisfies both the accuracy and real-time detection requirements.

Engineering Justification:

For a drone vision system, both high accuracy and low latency are important. The system must detect objects quickly while maintaining reliable detection performance. Model C provides a suitable balance between accuracy and real-time performance. Its lightweight computational requirement should also be verified on the actual drone hardware before deployment.

Final Solution:

Model C should be selected because it satisfies the specified accuracy and real-time detection constraints.

Accuracy = 86% $\geq$ 85%

Latency = 55 ms $<$ 60 ms

Conclusion:

Model C is the most suitable model for the drone vision system because it satisfies the accuracy requirement, meets the real-time latency constraint, and provides the best overall balance for lightweight drone deployment.

3. Security System

Recognition A security system must:

- High accuracy ($>92\%$)
- Moderate computation Options:

Traditional: 85%, low computation

Deep Learning: 94%, high computation

Hybrid: 91%, moderate computation

Compare and evaluate the best approach. Justify logically.

A. Security System Recognition

Problem:

A security recognition system requires high accuracy of more than 92% and moderate computational requirements. Three approaches are available: Traditional with 85% accuracy and low computation, Deep Learning with 94% accuracy and high computation, and Hybrid with 91% accuracy and moderate computation.

Comparison and Analysis:

The Traditional approach requires low computation, but its accuracy is only 85%, which is below the required 92%. Therefore, it does not satisfy the accuracy requirement.

The Deep Learning approach provides 94% accuracy, which satisfies the required accuracy of more than 92%. However, it requires high computational resources, which does not match the moderate computation requirement.

The Hybrid approach requires moderate computation, which satisfies the computational requirement. However, its accuracy is only 91%, which is below the required 92%. Therefore, it also fails to satisfy all requirements.

Engineering Justification:

The problem presents a trade-off between accuracy and computational cost. The Deep Learning approach is the only method that satisfies the accuracy requirement, but its high computational cost violates the moderate computation constraint. The Hybrid approach satisfies the computation requirement but fails to achieve the required accuracy.

Final Solution:

Based strictly on the given constraints, none of the three approaches completely satisfies both requirements. If high accuracy is the primary requirement, Deep Learning should be selected because it achieves 94% accuracy. However, it would require optimization techniques such as model compression, pruning, or quantization to reduce its computational requirements to a moderate level.

Conclusion:

Deep Learning is the best starting approach because it is the only method achieving accuracy above 92%. However, it must be optimized to meet the moderate-computation requirement.

4. Motion Tracking in Sports Analytics

Requirements:

- Accurate motion tracking
- Real-time performance Options:

Optical Flow: accurate, slow

Motion Estimation: fast, less accurate

Compare and evaluate the suitable approach. Justify using constraints.

A. Motion Tracking in Sports Analytics

Problem:

A sports analytics system requires accurate motion tracking and real-time performance. Two approaches are available: Optical Flow, which provides accurate tracking but is slow, and Motion Estimation, which is fast but less accurate.

Comparison and Analysis:

Optical Flow provides accurate motion tracking, which is important for analyzing player and ball movements. However, its slow processing speed makes it less suitable for real-time sports analytics.

Motion Estimation provides faster processing and is suitable for real-time applications. However, its lower accuracy may affect the precision of motion tracking and analysis.

Engineering Justification:

The system requires both accuracy and real-time performance. Therefore, neither method perfectly satisfies both requirements. Since real-time performance is an explicit requirement, Motion Estimation is the more suitable choice. Its speed allows the system to track movements immediately, although its accuracy may need improvement.

Final Solution:

Motion Estimation should be selected for real-time sports analytics because it provides fast motion tracking. Its accuracy can be improved using better feature extraction, filtering, or combining it with selected Optical Flow techniques.

Conclusion:

Motion Estimation is the preferred approach when real-time performance is critical. Optical Flow should be preferred when tracking accuracy is more important than processing speed.

5. Face Recognition System Design Constraints:

- Accuracy $\geq 90\%$
- Works in low light
- Moderate computation

Options:

Traditional: 82%

Deep Learning: 93%

Hybrid: 89%

Compare and evaluate the best choice. Justify logically.

A. Face Recognition System Design

Problem:

A face recognition system requires accuracy of at least 90%, reliable performance in low-light conditions, and moderate computational

requirements. Three approaches are available: Traditional with 82% accuracy, Deep Learning with 93% accuracy, and Hybrid with 89% accuracy.

Comparison and Analysis:

The Traditional approach provides only 82% accuracy, which is below the required 90%. Therefore, it does not satisfy the accuracy requirement and is not suitable.

The Deep Learning approach provides 93% accuracy, which satisfies the required accuracy of at least 90%. Deep Learning can also be trained and enhanced using low-light image datasets and suitable preprocessing techniques. However, its computational requirement may be high, so optimization is required to achieve moderate computation.

The Hybrid approach provides 89% accuracy, which is below the required 90%. Although it may offer a better balance of computational cost, it fails to satisfy the accuracy constraint.

Engineering Justification:

Accuracy is a critical requirement in face recognition because incorrect identification can reduce system reliability. Among the available approaches, Deep Learning is the only one that satisfies the accuracy requirement. It can also be adapted for low-light conditions using image enhancement and training with diverse low-light data. Model optimization techniques such as pruning, quantization, or a lightweight network can be used to reduce computational requirements.

Final Solution:

Deep Learning should be selected because it provides the highest accuracy of 93% and is the only approach that satisfies the accuracy requirement. It should be optimized and trained with low-light data to satisfy the remaining system constraints.

Conclusion:

Deep Learning is the best choice for the face recognition system because it provides 93% accuracy, supports adaptation to low-light conditions, and can be optimized to achieve moderate computational requirements.